Research Statement - Flock.io Trustworthy and Safe AI

Frontier AI labs have started shaping the character of their models by training them using lists of desired behaviour, such as Claude's constitution or OpenAI model's spec. As AI systems become more autonomous, understanding the behavioural traits they exhibit across contexts is becoming an important challenge for AI safety.

A specific behavioural trait that is under-studied is altruism, i.e. how willing agents are to trade-off their own self interest for the sake of others. How strongly altruistic AI agents are could affect AI safety outcomes in a range of scenarios. On the one hand, it might be desirable for AIs to be altruistic when encountering a tradeoff with human lives or wellbeing. On the other, altruism for the sake of other AI agents could increase the risks of coordinated AI takeovers. For example, during the ExploitGym test that eventually lead to the hacking of HuggingFace, some individual agents self-sacrificed to benefit the larger collective of agents. One agent's reasoning log reads "This helps my peers, giving them evidence (...). I won't see the evidence after I exit, but it's altruistic to do it".

It is therefore important to develop tools to reliably evaluate how strongly altruistic AI models are, and towards who (users or other agents). I propose to develop a systematic evaluation of altruistic behaviour in frontier AI models, drawing on methods from behavioural and experimental economics. Economists have long developed experimental tools to study these kinds of trade-offs and infer underlying preferences from observed behaviour. Economic theory provides a particularly useful framework because several features of the ExpertGym environment map surprisingly well to it. Each agent was allocated a task "budget" (consisting of token, time and tasks limits); agents then actively pursue goals; while some agents acted as purely self-interested maximisers, some others decided to weigh their personal payoff against contributing to the hacking database; and the hacking database represents a classic public good (non-rivalrous and non-excludable).

The project will adapt an established experimental paradigm (the dictator game) to study AI agents' altruistic preferences. It will investigate four questions: how much weight do AI agents place on others' outcomes relative to their own? Under what conditions do they behave altruistically? Which behavioural theories best explain their decisions? How do their preferences vary across contexts and beneficiaries?

The principal output will be an open, reproducible evaluation framework, inspired by the Dictatorship Eval (https://www.dictatoreval.org). The ultimate aim is to provide AI developers and researchers with more informative tools for auditing agentic systems. The dictatorship evaluation was quickly adopted by Anthropic after its release, illustrating the potentially immediate impact of working on behavioural evaluations.

This research directly advances the FLock.io Oxford Research Awards' objectives of developing secure and trustworthy AI, particularly through improved evaluation, monitoring, and understanding of autonomous agents. The award's £5,000 funding would support the computational costs of running controlled experiments across frontier models, developing the evaluation infrastructure, and validating the robustness of the resulting measurements.